

Best Approach for Storing Bill Payments in Prisma

1. Key Design Points

- Every bill payment is a transaction.
 - Transaction: the generic money movement record.
 - BillPayment: a child record with bill-specific info.
- This ensures you don't pollute the Transaction table with airtime/data/PHCN-specific fields.

2. Two-step saving process

When a user pays a bill:

- First, create a Transaction with:
 - type = BILL_PAYMENT (or VTU_PURCHASE)
 - status = PENDING
 - amount = request.amount
- Then create a linked BillPayment record:
 - transactionId = transaction.id
 - provider = MTN | GLO | PHCN | DSTV etc.
 - status = PENDING

Once VTPass confirms the transaction, update both statuses -> COMPLETED or FAILED.

3. Why this is good

- Consistency: all money movements live in Transaction (whether deposit, transfer, or bill payment).
- Flexibility: bill-specific metadata stays in BillPayment.
- Auditing: AuditLog and FraudReport can tie back to both if needed.

Workflow Example (Bill Payment Flow)

1. User buys 500 NGN MTN airtime.

2. Backend creates a Transaction:

```
const transaction = await prisma.transaction.create({  
  data: {  
    userId,  
    walletId,  
    type: "BILL_PAYMENT",  
    amount: 500,  
    status: "PENDING",  
    reference: generateReference(),  
  }  
});
```

3. Then a BillPayment linked to that transaction:

```
await prisma.billPayment.create({  
  data: {  
    userId,  
    transactionId: transaction.id,  
    provider: "MTN",  
    amount: 500,  
    status: "PENDING",  
  }  
});
```

4. Send request to VTPass.

5. When response comes back -> update both:

```
await prisma.transaction.update({  
  where: { id: transaction.id },  
  data: { status: "COMPLETED" }  
});
```

```
await prisma.billPayment.update({  
  where: { transactionId: transaction.id },  
  data: { status: "SUCCESS" }  
});
```

Reporting Benefits

- You can query all bill payments easily:

```
SELECT * FROM BillPayment WHERE provider = 'MTN' AND status = 'SUCCESS';
```

- Or query user's transactions across all activities:

```
SELECT * FROM Transaction WHERE userId = '...';
```

Summary:

- Keep bill payments as a child of transactions.
- Always create Transaction first, then BillPayment.
- Update both statuses when you get callback/confirmation.